



LECTURE 3

Understanding User and Stakeholder Needs

COEN / ELEC 390

Acknowledgement

- The content of these slides is based on Chapters 8 and 11 of the textbook :

Product Design and Development by K.T. Ulrich, S.D. Eppinger
McGraw Hill Irwin

Requirements vs Specifications

- **Requirements** (sometimes called user requirements) are what the user wants the product to do, or a property the user wants the product to have.
- **Specifications** are for the developer/designer that tell them what needs to be built and whether it is built correctly
 - Usually specifications are linked to a requirement
 - Once we have fully specified a requirement we usually have a series of yes/no questions. If the answer to all questions is correct the requirement is correctly built.
 - The yes/no questions are called acceptance tests
 - Key idea: specifications are for the developer/designer

Eliciting requirements

- Understanding the user needs is a challenging problem
- Three syndromes that complicate it
 - “yes, but”:
 - For the first time after months or years or waiting-they have the opportunity to interact with the system. And guess what: it’s not exactly what they expected! “I like it yes, but It would be nice if ...”
 - “Undiscovered ruin”:
 - The more you find, the more you know remains. It is hard to know when you are done requirement elicitation
 - “user and developer”:
 - Communication gap between user and developer
 - Few application developers have any training in any elicitation techniques

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Problem	Solution
Users do not know what they want, or they know what they want but cannot articulate it.	Recognize and appreciate the user as domain expert
Users think they know what they want until developers give them what they said they wanted	Provide alternative elicitation techniques earlier: storyboarding, role playing, and so on.
Analysts think they understand user problems better than users do.	Put the analyst in the user's place. Try role playing for an hour or a day.
Everybody believes everybody else is politically motivated.	Yes, its part of human nature, so let's get on with the program.



Interviewing

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Introduction

- Interviews are a simple and direct approach to understand the user's needs in most situations.
- Beware of the user and developer syndrome (communication gap between user and developer).
- One of the key goals of interviewing is to make sure that the biases and predispositions of the interviewer do not interfere with a free exchange of information -Not so easy.

Context-Free Questions

- Question should not contain context of potential solution. If not we, may bias the response we get
- These questions force us to listen before attempting to invent or describe a potential solution
- Examples:
 - Who is the user?
 - Who is the customer? (economic buyer)
 - Are their needs different?
 - Where else can a solution to this problem be found?
 - What would you like?
 - What are some things that bother you?

Solutions-Context Questions

- We should explore solutions after the context-free questions have been asked and answered.
- Give the customer information about your current solutions and elicit more information from them.

The Moment of Truth: The Interview

- Normally just one person conducts the interview, asking questions and recording answers.
- For your project we require at least 2 interviews. All members should attend both interviews. If you do more, not all must attend.
- Best to get a team member who is engaging to conduct the interview(i.e. do the talking). This person's job is to **animate** the interview, to get the customer talking.
- Ideally have one other who intervenes rarely, but who can if the interviewer misses something. This person job is to **listen** and to very occasionally intervene.
- Have at least one other's whose job is to **record** or take notes in the interview. It would be better to have two people doing this. These people should not talk, but only listen and make notes.

The Moment of Truth: The Interview

- Tips for a successful interview:
 - Prepare an appropriate context-free interview. Write it down in notebook.
 - Before the interview, research the background of the stakeholder and the company to be interviewed.
 - Write down the answers in your notebook during the interview.
 - Refer to the template during the interview to make certain that you're asking the right questions.
- Your script should not be too constraining.
- If your interviewee slips into a riff where they are describing their needs and problems you should be very happy. Let them talk and write down their answers.
- At important points you may want to ask follow up questions
 - These cannot be predicted. They depend on what the user said to you.

Interview Template

Source: Product Design and Development by K.T. Ulrich, S.D. Eppinger

- Part 0: Set the interviewee at ease. Once rapport has been established, the interview is likely to take on a life of its own.
- Part I: Establishing the customer or user Profile
- Part IIA: Assessing the problem
- Part IIB: Understanding the user environment
- Part IIC: Recap for understanding
- Part III: The analyst's inputs on the customer problem
- Part IV: Assessing your Solution (if applicable)
- Part V: Assessing the opportunity (who will use it? How valuable?)
- Part VI: Assessing the reliability, performance and support needs
- Part VII: Other requirements
- Part VIII: Wrap-up
- Part IX: The Analyst's Summary

Continued

- Part 0: Set interviewee at ease
 - To help the interviewee relax, open the interviewee with some “small talk” (weather, any trouble getting here, etc).
 - Thank the interviewee.
- Part I: Establishing the Customer or User Profile
 - Adopt to your area. If you are making something that will be sold to a company you will want details on the person’s position in the organization, what they do in their work. If you are making something for consumers you may want to identify their interests, where and under what conditions they might use your product.

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- **Part IIA: Assessing the Problem**

- *This part is the “blue sky” part of the interview. Mention only the general domain, not what you think are problems for this domain, or what you see as solutions for the problems you have.*
- For which *[application type]* problems do you lack good solutions?
- What are they? (*Hint: Keep asking, “Anything else?”*)
- *For each problem, ask the following questions.*
 - *Why does this problem exist?*
 - *How do you solve it now?*
 - *How would you like to solve it?*

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- **Part IIB: Understanding the User Environment**

- Who are the users?
- Which platforms are in use?
- Where will they use the app?
- Are additional applications in use that are relevant to this application?

- **Part IIC: Recap for Understanding**

- You have told me:
- *(List customer-described problems in your own words.)*
- Does this adequately represent the problems you are having with your existing solution?
- What, if any, other problems are you experiencing?

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- **Part III: The Analyst's Inputs on the Customer's Problem**

- Now start to tell the interviewee about some of the problems that you have thought about for this domain. Don't tell them your proposed solutions. If they already mentioned in Part II one of the problems you had pre-identified, you don't need to repeat it here. List any needs or additional problems you think should concern the customer or user.)
- For each suggested problem, ask the following questions.
 - Is this a real problem?
 - What are the reasons for this problem?
 - How do you currently solve the problem?
 - How would you like to solve the problem?
 - How would you rank solving these problems in comparison to others you've mentioned?

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- **Part IV: Assessing Your Solution *(if applicable)***
 - *Now tell the interviewee what your product will do for them, how it will solve the problems identified before.*
 - *(Summarize the key capabilities of your proposed solution.)*
 - What if you could:
 - Do A
 - Do B
- How would you rank the importance of these? Ask them to explain.

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- **Part V: Assessing the Opportunity**

- Who in your organization needs this application?
- How would you value a successful solution?
- Is there another part of your life where you could use this application?
- What price would you pay for a product that had this capability?

- **Part VI: Assessing the Reliability, Performance, and Support Needs**

- What are your expectations for reliability?
- What are your expectation for performance?

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- **Part VII: Other Requirements**

- Are there any legal, regulatory, or environmental requirements or other standards that must be supported?
- Can you think of any other requirements we should know about?

- **Part VIII: Wrap-up**

- Are there any other questions I should be asking you?
- If I need to ask follow-up questions, may I give you a call? Would you be willing to participate in a requirements review?
- Thank the interviewee.

- **The analyst's summary (Part XI)**

- Your three most important needs/problems, look for common needs
- Ten interviews will often create 10-15 different needs. This is the start of your requirements repository

A Note on Questionnaires

- The team cannot substitute a questionnaire for the interviewing process.
- Indeed, some have concluded that the questionnaire technique suppresses almost everything good about requirements gathering.
- If the applications has a large number of existing or potential users and if the goal is to provide statistical input about user or customer preferences among a limited set of choices, a questionnaire can be used.